The study of chromosomes, their structure, and their inheritance is called **cytogenetics**. The science of human cytogenetics dates from 1956, when it was first established that the normal human chromosome number is 46. Since that time, much has been learned about human chromosomes, their normal structure and composition, and the identity of the genes that they contain, as well as their numerous and varied abnormalities.

With the exception of cells that develop into gametes (the **germline**), all cells that contribute to one's body are called **somatic cells** (*soma,* body). The genome contained in the nucleus of human somatic cells consists of 46 chromosomes, made up of 24 different types and arranged in 23 pairs (Fig. 2-1). Of those 23 pairs, 22 are alike in males and females and are called **autosomes,** originally numbered in order of their apparent size from the largest to the smallest. The remaining pair comprises the two different types of **sex chromosomes:** an X and a Y chromosome in males and two X chromosomes in females. Central to the concept of the human genome, each chromosome carries a different subset of genes that are arranged linearly along its DNA. Members of a pair of chromosomes (referred to as **homologous chromosomes** or **homologues**) carry matching genetic information; that is, they typically have the same genes in the same order. At any specific locus, however, the homologues either may be identical or may vary slightly in sequence; these different forms of a gene are called **alleles**. One member of each pair of chromosomes is inherited from the father, the other from the mother. Normally, the members of a pair of autosomes are microscopically indistinguishable from each other. In females, the sex chromosomes, the two **X chromosomes,** are likewise largely indistinguishable. In males, however, the sex chromosomes differ. One is an X, identical to the Xs of the female, inherited by a male from his mother and transmitted to his daughters; the other, the **Y chromosome,** is inherited from his father and transmitted to his sons. In Chapter 6, as we explore the chromosomal and genomic basis of disease, we will look at some exceptions to the simple and almost universal rule that human females are XX and human males are XY.

In addition to the nuclear genome, a small but important part of the human genome resides in mitochondria in the cytoplasm (see Fig. 2-1). The mitochondrial chromosome, to be described later in this chapter, has a number of unusual features that distinguish it from the rest of the human genome.

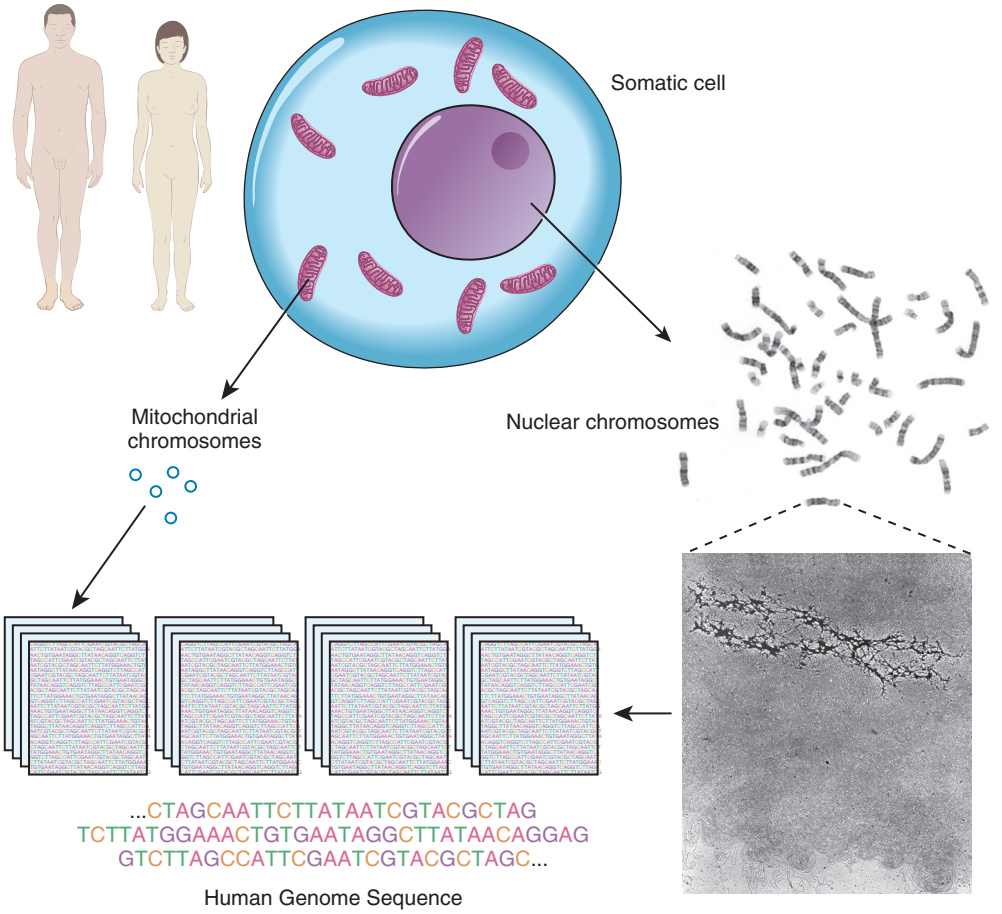


Figure 2-1 The human genome, encoded on both nuclear and mitochondrial chromosomes. *See Sources & Acknowledgments.*